ARTIFICIAL INTELLIGENCE (AI) USAGE AND CONTRIBUTION DECLARATION 5 BCA SPECIALIZATION PROJECT

1. Project Details

Project Title:	Autonomous Video Ad-Placement and Contextual Safety Analyzer (AdSafe Studio)
Programme:	Bachelor of Computer Applications (BCA) — BCA482-5, Specialization Project
Semester / Section:	V / 5BCA-B
Academic Year:	2026–27
Project Guide:	Dr. Chanti.S, Assistant Professor, Department of Computer Science

2. AI Usage Declaration

We, the undersigned students, declare that Artificial Intelligence (AI) tools, wherever used during the preparation and development of this project, were used as supportive tools with academic responsibility.

We have disclosed the AI tools used and the nature of their contribution below. We understand that the use of AI tools does not replace our responsibility for the academic, technical, and intellectual work submitted as part of this project.

All AI-assisted content, code, suggestions, or analysis incorporated into the project has been reviewed, verified, tested, and appropriately modified by us.

2.1 AI Tools Used

S No.	AI Tool	Purpose of Use	Project Section / Stage
1	Claude (Anthropic) Opus 5 and Opus 4.8, used through the Claude Code CLI	Code generation and refactoring; debugging; algorithm and methodology suggestions; drafting of the report, presentation and project documentation; generation of diagrams and figures.	Implementation of all four modules; Stage 5 report; presentation deck; supporting documentation. Nearly every commit in the project repository records an AI co-author.
2			
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Rows 2–4 are left blank for any further tool used by an individual team member. faster-whisper and Ultralytics YOLOv8 are not listed above: they are pretrained models that form part of the delivered system itself rather than authoring aids, and they are documented in the implementation chapter of the report.

2.2 Nature of AI Assistance

Tick or mark the applicable options and provide a brief description where necessary.

Type of Assistance	Used	Brief Description
Concept clarification / learning	**Yes**	Behaviour of YOLOv8 and Whisper; SQLite WAL journaling and foreign-key enforcement; CSS flex and grid sizing rules.
Brainstorming / idea generation	**Yes**	Approaches for locating advertisement surfaces, and for distinguishing rendered game content from filmed content.
Literature / research assistance	**Yes**	Sourcing published demonetization statistics used to justify the problem; verifying which risk classes the Open Images V7 taxonomy can actually supply.
Programming / code generation	**Yes**	Source code across the backend, frontend and AI pipeline was drafted with AI assistance and then reviewed, tested and modified by us.
Code debugging	**Yes**	Startup traceback caused by a NumPy 2 ABI mismatch; partial data loss on demo reset; missed weapon detections on real footage; horizontal layout overflow; a race that suppressed the progress bar.
Algorithm / methodology suggestions	**Yes**	Maximal-rectangle search for free advertisement space; temporal-variance detection of persistent overlays; multi-scale inference for objects at differing distances.
Data analysis / interpretation	**Yes**	Interpreting mAP@50 against observed behaviour on real video, which led us to reject fine-tuning run 2 despite its better validation scores.
Report writing / documentation	**Yes**	Drafting the Stage 5 report in the prescribed format, and the README, roadmap and project record.
Grammar / language improvement	**Yes**	Editing and rephrasing throughout the report and the presentation.
Images / diagrams	**Yes**	All twelve system-model and project-planning diagrams were generated programmatically with matplotlib; the interface screenshots are captured by a script.
Other	**Yes**	Generation of the presentation deck; automation of screenshot capture and of layout regression checks.

3. Student Responsibility and Authenticity

We confirm that:

1. We have reviewed and verified all AI-assisted content incorporated into the project.
2. We understand and can explain the concepts, methodology, implementation, code, analysis, results, and conclusions presented in the project.
3. We have not knowingly submitted AI-generated work as our original work without appropriate review or acknowledgement.
4. AI tools have not been used to fabricate or manipulate data, results, observations, references, or research evidence.
5. Any AI-generated code used in the project has been tested, understood, and modified where necessary.
6. We accept full responsibility for the accuracy, originality, and academic integrity of the submitted work.

3.1 Significant AI Interaction Record

Complete this section when AI has been substantially used for code, methodology, analysis, content generation, or other project development activities.

Date	AI Tool	Purpose / Query	How the Output Was Used
22 Jul 2026	Claude Code	Scaffold the application: FastAPI backend, React interface, SQLite schema, with a deterministic stand-in for each AI engine.	Generated code was read, run and extended by us; retained as the project baseline.
01 Aug 2026	Claude Code	Replace the stand-in transcriber with faster-whisper, including word-level timestamps required by the AudioFlags schema.	Integrated after we tested it against sample audio and confirmed the timestamps aligned.
02–03 Aug 2026	Claude Code	Prepare an Open Images V7 subset and fine-tune YOLOv8 for the project's risk classes.	We ran the training ourselves. Run 1 was accepted (mAP@50 0.566); run 2 was rejected after we tested it on real video.
03 Aug 2026	Claude Code	Design and implement the free-space advertisement placement algorithm.	Implemented, then verified by us against sample footage before acceptance.
07 Aug 2026	Claude Code	Diagnose why a firearm visible in real footage was not detected, and propose a correction.	Multi-scale inference adopted only after we confirmed it worked on both a distant and a close-up clip.
07 Aug 2026	Claude Code	Detect persistent on-screen overlays so that weapons rendered inside a game are weighed differently from filmed ones.	We tested it on real gameplay footage; the stability threshold was changed to a relative one because the first version missed real HUDs.
16–17 Aug 2026	Claude Code	Draft the Stage 5 report in the prescribed LaTeX format and generate the system-model diagrams.	Reviewed and corrected by us; several diagrams were regenerated after we found geometry errors on inspection.
20 Aug 2026	Claude Code	Automate capture of the interface screenshots, and correct the defects that the automation exposed.	Fixes reviewed and accepted after the full test suite (169 tests) was re-run and passed.

Note: The Project Guide or Evaluation Committee may request supporting AI interaction records where necessary. The full record is preserved in the project's version control history, in which nearly every commit carries an AI co-author trailer and each commit message states what was changed and why. The table above lists the most significant interactions, not every one.

Student Declaration

We certify that the information provided in this document is true and complete to the best of our knowledge. We understand that failure to disclose significant AI usage or misrepresentation of AI-generated work may be treated as a violation of academic integrity.

No.	Student Name	Register Number	Signature
1	Andrew Shelton S	2441609	
2	Akash Ganesh	2441604	
3	Andrew Judah	2441608	
4	Ramona Elisha Mary	2441646	
5			

Date: _____ Place: Bengaluru, Karnataka

Project Guide

Name: Dr. Chanti.S

Signature: _____

Project Coordinator

Name: _____

Signature: _____